

Prevalence of depression and its associated factors among people living with HIV in Nepal: Systematic Review

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Citation

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REVIEW TITLE AND BASIC DETAILS

Review title

Prevalence of depression and its associated factors among people living with HIV in Nepal: Systematic Review

Condition or domain being studied

Depression; HIV infection; Depressive symptoms; HIV Status ; Major depressive disorder; Mental disorder

Rationale for the review

The prevalence of depression among the 15-49 year old Nepalese population is about 2% to 5% according to a recent NDHS 2022 survey (2). However, people living with HIV (PLHIV) face a substantially higher risk of depression compared to the general population, being approximately twice as likely to experience major depressive symptoms (3). At the global level, the prevalence of depression among PLHIV is estimated to be around 31% (4). In Nepal, however, studies report considerable variability, with prevalence rates ranging from 10% to 43% (5,6). Comparable evidence from India indicates a prevalence of 39.1% (95% CI: 30.4–48.5), with even higher rates of approximately 45.5% reported in North India, a region with geographical similarities to Nepal (7). Depression among PLHIV is influenced by a range of interrelated factors. Previous studies have identified socioeconomic factors such as education, employment, and marital status; psychological and behavioral factors including substance use, family support, and internalized stigma; and clinical factors such as CD4 count and duration of antiretroviral therapy (ART) as key determinants (8–11).

In Nepal, stigma and discrimination against PLHIV remain highly prevalent, rooted in cultural norms, moral judgments, gender inequality, and structural factors (12). Previous study showed about 70% prevalence of stigma and 34% prevalence of discrimination (13). These experiences of stigma and discrimination are strongly associated with an increased risk of depression among this population (7,10). This review will be the first comprehensive synthesis of evidence on depression among PLHIV in Nepal, aims to generate robust empirical evidence to inform evidence-based health decision-making.

Review objectives

Objectives

The primary objective of this review is to systematically search all existing evidence on the prevalence of depression among people living with HIV in Nepal and do the relevant synthesis of the evidence.

The secondary objectives are to:

1. Identify and synthesize all existing literature reporting depression prevalence among PLHIV in Nepal.
2. To identify the pooled prevalence rate from all possible studies.
3. Assess the risk of bias and methodological quality of included studies.
4. Explore sources of heterogeneity through subgroup and sensitivity analyses.
5. Explore the factors associated with the occurrence of depression.

Review Question

What are the all available studies of depression in PLHIV in Nepal and what is the pooled prevalence of depression among PLHIV in Nepal, and what are the associated socio-demographic and clinical factors?

Keywords

Systematic review; Meta-analysis; Mental disorders; Mental health; Depression; Depressive disorder; HIV; People living with HIV; HIV/AIDS; Nepal

Country

Nepal

ELIGIBILITY CRITERIA

Population

Included

People living with HIV in Nepal, regardless of age, antiretroviral therapy (ART) status, or care setting (community-based, institutional, or clinic-based).

Intervention(s) or exposure(s)

Included

HIV testing

Confirmed HIV infection participants.

Comparator(s) or control(s)

This review does not have any comparators

Study design

Both randomized and nonrandomized study types will be included.

Included

Observational studies, including cross-sectional, case-control, and cohort studies, and interventional studies (randomized or non-randomized) that report baseline (pre-intervention) prevalence or data on depression among people living with HIV in Nepal.

Context

Studies in People living with HIV in Nepal, regardless of age, antiretroviral therapy (ART) status, or care setting (community-based, institutional, or clinic-based).

TIMELINE OF THE REVIEW

Date of first submission to PROSPERO

01 May 2026

Review timeline

Start date: 1 February 2001. End date: 26 July 2026.

Date of registration in PROSPERO

02 May 2026

AVAILABILITY OF FULL PROTOCOL

Availability of full protocol

A full protocol has been written and uploaded to PROSPERO. The protocol may be accessed through this link

<https://www.crd.york.ac.uk/PROSPEROFILES/ff4e6438382857de078a584a14c515fd.pdf>.

SEARCHING AND SCREENING

Search for unpublished studies

Only published studies will be sought.

Main sources that will be searched

The main databases to be searched are *Embase - Embase via Ovid, MEDLINE*.

Other important or specialist databases that will be searched

APA PsycINFO (via OVID)

NepMed (Nepal Medline)

WHO Global Index Medicus.

Google Scholar

Search language restrictions

There are no language restrictions.

Search date restrictions

There are no search date restrictions.

Other methods of identifying studies

Other studies will be identified by: *contacting authors or experts, looking through all the articles that cite the papers included in the review ("snowballing" or forward citation searching), reference list checking (backward citation searching) and searching dissertation and thesis databases.*

Link to search strategy

A full search strategy is available in the full protocol as described in the *Availability of full protocol* section

Selection process

Studies will be screened independently by at least two people (or person/machine combination) with a process to resolve differences.

DATA COLLECTION PROCESS

Data extraction from published articles and reports

Data will be extracted independently by at least two people (or person/machine combination) with a process to resolve differences.

Authors will be asked to provide any required data not available in published reports.

Study risk of bias or quality assessment

Risk of bias will be assessed using: *Newcastle-Ottawa*

Joanna Briggs Institute (JBI)

Data will be assessed independently by at least two people (or person/machine combination) with a process to resolve differences.

Additional information will be sought from study investigators if required information is unclear or unavailable in the study publications/reports.

Reporting bias assessment

Risk of bias due to missing results will be assessed

Certainty assessment

The overall certainty of evidence for the primary outcome will be assessed using the adapted GRADE framework for prevalence estimates as described by Zhang et al. Certainty will be rated as high, moderate, low, or very low based on five domains: risk of bias, inconsistency, indirectness, imprecision, and publication bias. Upgrading factors (e.g., large magnitude of effect, dose-response gradient) will not be applied as this review estimates a single-group prevalence rather than comparing interventions or exposures .

OUTCOMES TO BE ANALYSED

Main outcomes

The primary outcomes of this review will be descriptive analysis of selected studies and narrative synthesis or the pool prevalence of depression in the study population based on available relevant studies.

Additional outcomes

Subgroup analyses of depression prevalence in the study population, along with meta-regression based on relevant covariates, will be conducted as secondary outcomes of this review.

PLANNED DATA SYNTHESIS

Strategy for data synthesis

A narrative synthesis will be conducted for all included studies, and associated factors will be summarized narratively if meta-analysis is not feasible. Where sufficient data are available, a meta-analysis will be performed to estimate the pooled prevalence of depression. Quantitative analysis will be done in the R studio for meta-analysis, data summary and visualization. The Subgroup analyses will be conducted, and associated factors will be explored by pooling effect estimates (adjusted odds ratios or odds ratios, where appropriate). Statistical heterogeneity will be assessed using the I^2 statistic and the Chi^2 test (Cochrane Handbook, Section 10.10.2). I^2 will be interpreted with the following approximate thresholds; 0–40%: might not be important, 30–60%: moderate heterogeneity, 50–90%: substantial heterogeneity, 75–100%: considerable heterogeneity. However, these thresholds are not absolute; interpretation will also consider the magnitude and direction of effects and the Chi^2 p-value (significance set at $p < 0.10$).

Since our review is based on prevalence, we will expect high variability in study design, geographical region, sampling technique, depression assessment tools, etc. In such cases, a random-effects model will be used. If I^2 exceeds 90%, subgroup analysis and meta-regression will be done to explore heterogeneity.

CURRENT REVIEW STAGE

Stage of the review at this submission

Review stage	Started	Completed
Pilot work	✓	✓
Formal searching/study identification		
Screening search results against inclusion criteria		
Data extraction or receipt of IPD		
Risk of bias/quality assessment		
Data synthesis		

Review status

The review is currently planned or ongoing.

Publication of review results

Results of the review will be published in English.

REVIEW AFFILIATION, FUNDING AND PEER REVIEW

Review team members

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No conflict of interest declared.

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No conflict of interest declared.

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No conflict of interest declared.

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No conflict of interest declared.

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No conflict of interest declared.

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Funding source

Review has no funding and no agreed support from an academic institution and is done in authors' own time.

Peer review

There has been no peer review of this planned review.

ADDITIONAL INFORMATION

Review conflict of interest

Declared individual interests are recorded under team member details.. No additional interests are recorded for this review.

Medical Subject Headings

Depression; HIV Infections; Humans; Nepal; Prevalence; HIV; Mental Health; Mental Disorders; Asia, Southeastern; Systematic Review; Meta-Analysis; Review

SIMILAR REVIEWS

Check for similar records already in PROSPERO

PROSPERO identified a number of existing PROSPERO records that were similar to this one (last check made on 20 January 2026). These are shown below along with the reasons given by

that the review team for the reviews being different and/or proceeding.

- Prevalence of Depression among People Living with HIV/AIDS in India: A Systematic Review and MetaAnalysis [published 26 January 2024] [CRD42024498477]. The review was judged **not to be similar**
- Suicidal Ideation and Attempts Among People Living with HIV in India: A Systematic Review of Contributing Factors [published 10 January 2026] [CRD420261282417]. The review was judged **not to be similar**
- Prevalence of depression and anxiety among people living with leprosy and associated factors: a systematic review and meta-analysis [published 17 July 2024] [CRD42024566330]. The review was judged **not to be similar**
- Prevalence of Depression and Associated Factors among Adult People living with HIV/AIDS in India: A systematic review and meta-analysis [published 26 January 2024] [CRD42024502040]. The review was acknowledged as **similar** but the authors opted to continue because *there are differences in population, there are differences in intervention or comparator, the review will be more up to date, the review uses improved methods*

PROSPERO version history

- [Version 1.0, published 02 May 2026](#)

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